IMIC 2027

INTELLIGENT MANUFACTURING INTERNATIONAL CONTEST

DYNALOG





DYNALOG CONTEXT

- FIVES XCELLA's solution = Intralogistics and warehouse automation project
- Fleet of AGVs capable of navigating and accessing warehouse racks via elevators to retrieve or deposit packages

Characteristics

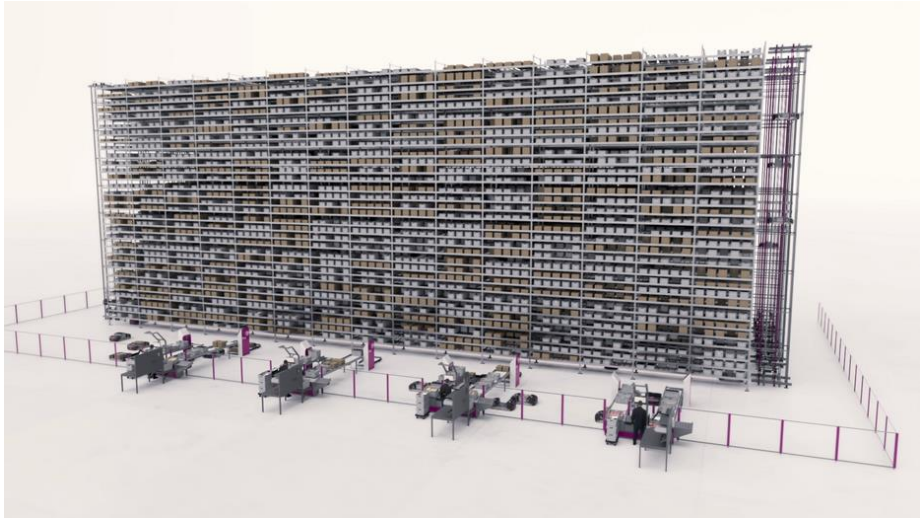
- High-density and vertical storage
- IN/OUT operations processed simultaneously
- Centralized software control
- Throughput: 1 bin per second
- AGV speed: 3 m/s on the ground



Video link: https://www.youtube.com/watch?v=A7SbZuiYZvM

WAREHOUSE DESCRIPTION

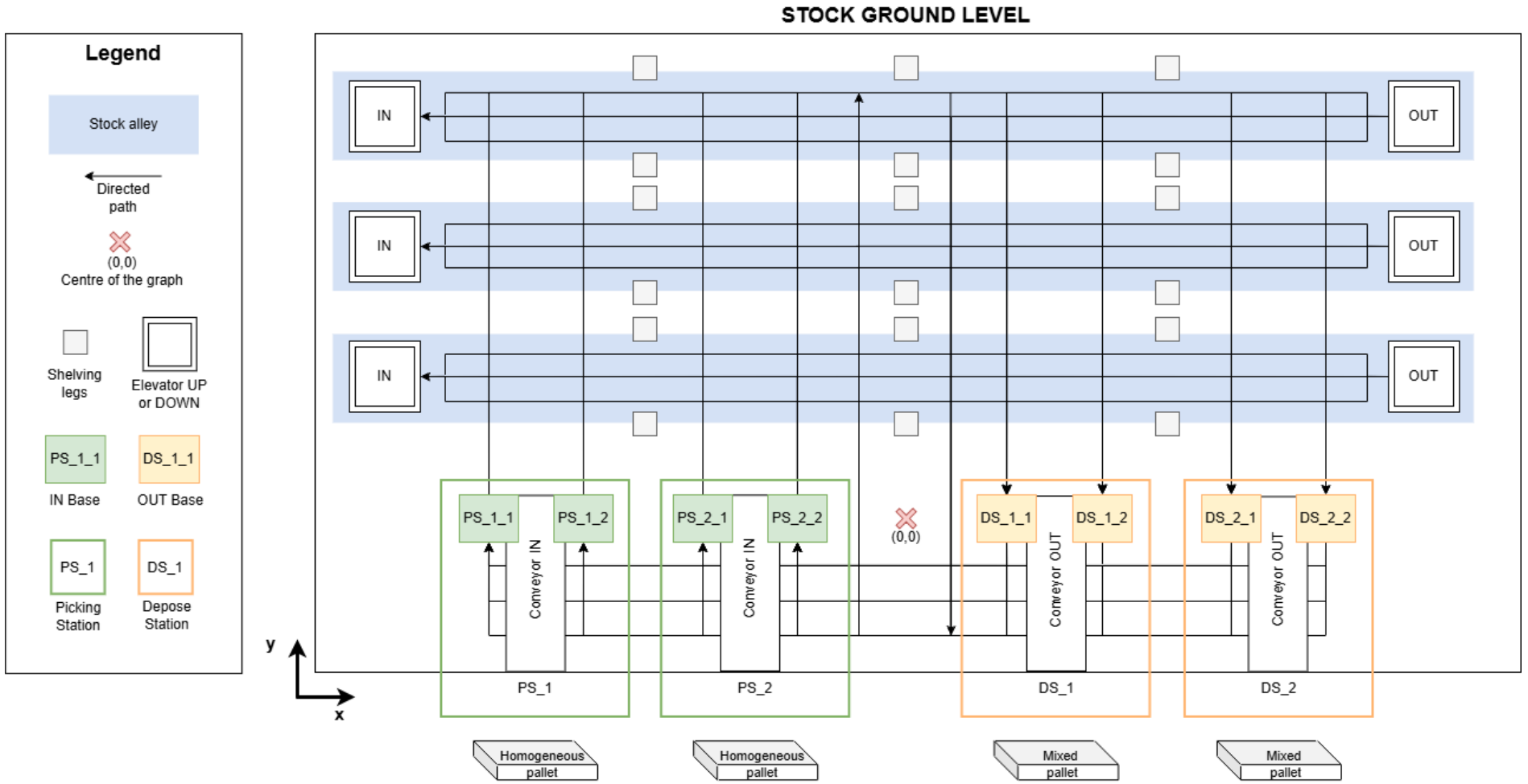
- Warehouse organized into **alleys** with **levels (= tubes)**
- **Elevators**: up on one side, down on the other
- Each tubes is divided in **bays**
- Each bay contains **10 storage locations** (5 left, 5 right)
- A floor space dedicated to pick and deposite packages: **IN and OUT Bases**
- Depalletizing robots feed **IN Bases** with packages coming from **homogeneous pallets**
- Palletizing robots retrieve packages from **OUT Bases** to complete **mixed pallets**
- The **AGVs** can travel on the floor, use elevators and travel in tubes



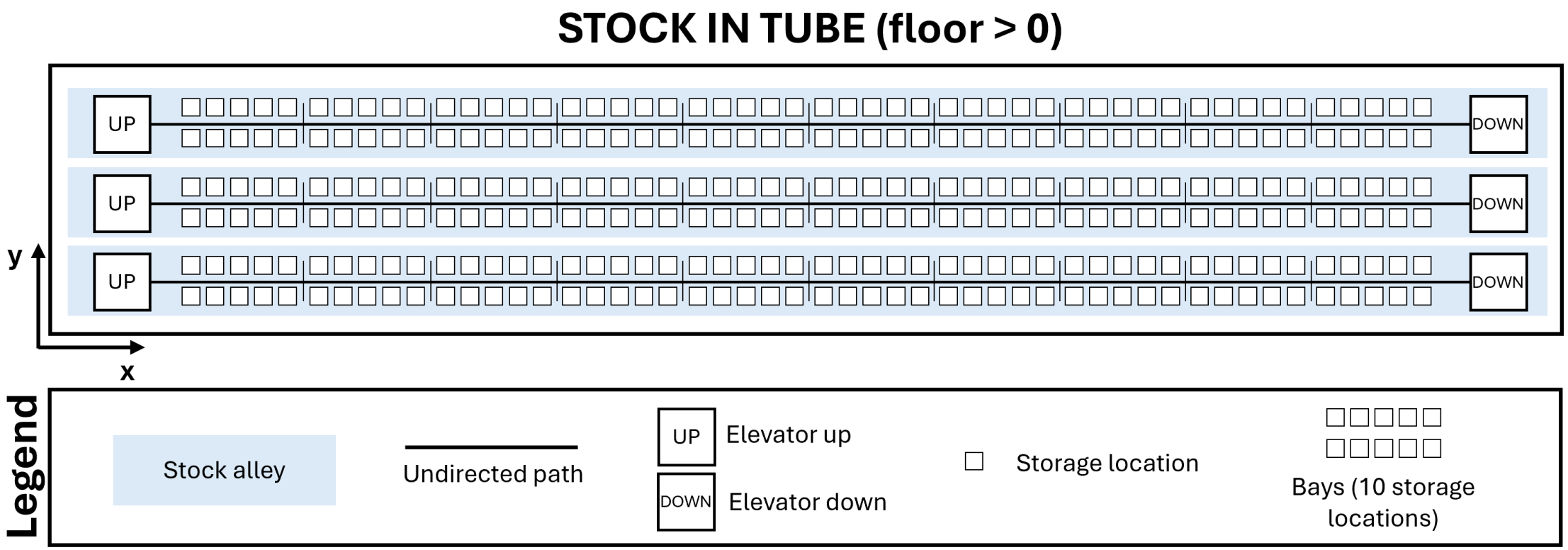


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WAREHOUSE DESCRIPTION



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GRAPH DESCRIPTION

- All movements permitted by the AGVs are modeled as a **directed graph**
- Each edge represents a path that an AGV can take to move through the warehouse. An edge can be:
  - directed (one direction) on the warehouse ground floor
  - undirected (both directions) in the tubes (storage levels)

- Each node represents a path intersection or a point of interest where an AGV can perform an ACTION:
  - **Picking up a package**: from IN Base or location in a tube
  - **Dropping off a package**: at OUT Base or location in a tube
  - **Attach / Detach** from an elevator
  - **Board** an elevator to go up to a tube / down to the ground floor
  - **Move** between two nodes in the graph
  - **Change direction**
  - **Wait** at a graph node

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IMIC WAREHOUSE CONFIGURATION

- 10 alleys and 10 tubes (levels) per alley
- 10 bays per alley, each with 5 locations on the right and 5 on the left
- 10 000 storage locations
- 1 Picking station (= 1 depalletizing robot): 4 IN Bases
- 2 Depose stations (= 2 palletizing robots): 4 OUT Bases
- Two sizes of AGV fleet are proposed: **40 AGV and 60 AGV**
- Filling rate of the stock: 80%

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MISSIONS

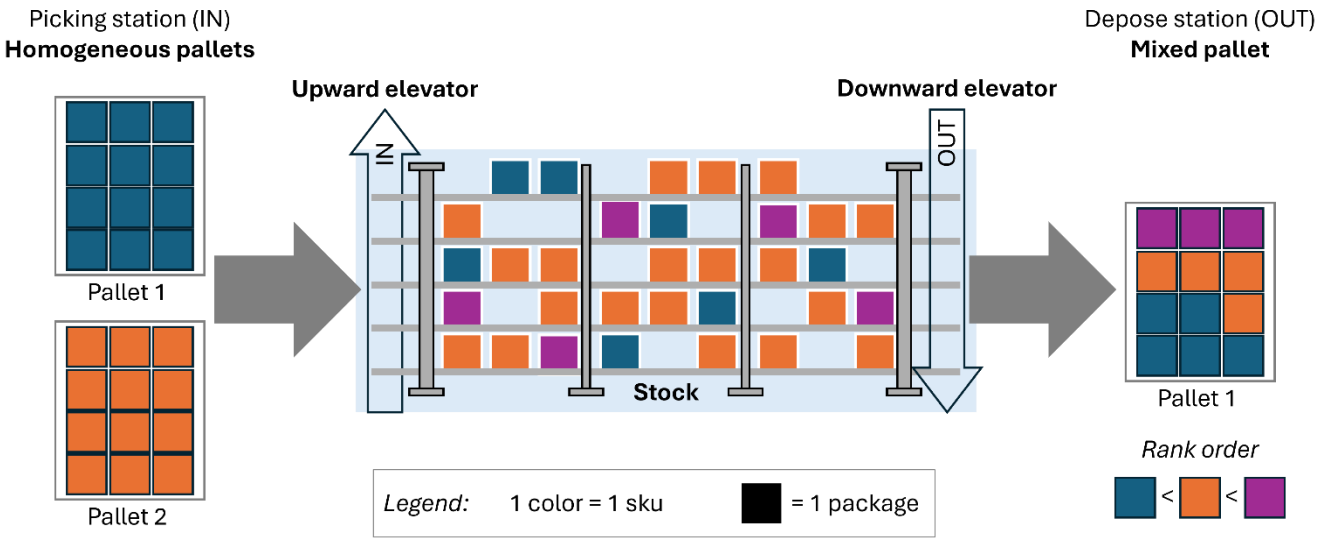
Two types of missions:

(steps for IN/OUT missions are listed and described on the Github)

- IN missions: storing an item in stock
- OUT missions: removing an item from stock in response to an order
- Consecutive IN and OUT missions = whenever is possible
- **/!\ Strict priority order** (**rank**) for filling the pallet (heavier items on the bottom, lighter items on top)

SYSTEM CONSTRAINTS AND FIXED PARAMETERS

- Limitations: 1 AGV/elevator
- **Redundancy**:
  - 1 homogeneous pallet = 1 SKU
  - 1 package = 1 SKU
  - But multiple stock locations with the same SKU
- **Rank**: Priority order of packages on OUT missions
  - Value from 0 (bottom) to 49 (top)
  - Reset to 0 for each new pallet
  - The rank MUST BE FOLLOWED
- Uniform pallet size = 50 packages



INPUTS FOR THE CONTESTANTS

- For each instance, the **list of IN/OUT missions over a 2-hour period**, in JSON format, including:
  - Initial stock: initial SKU and their locations
  - AGV: Initial positions
  - IN missions: SKU, date, IN Base where to pick up the package
  - OUT missions: SKU, date by which the pallet must be complete, OUT Base where to drop off the package, rank

- The impact of two parameters will be studied:
  - The size of AGV fleet
  - The coefficient of popularity

  } x5 instances

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DECISIONS AND KPI

The model developed must make the following **decisions**:

- Where to store (IN)? Where to retrieve (OUT)?
- Which AGV for which mission?
- In what order should the missions be executed?

KPIs:

- Global makespan (1) : from the first mission asked to the last one completed
- Makespan/pallet IN (2) : from the first package depalletized on the conveyor to the last one leaving the IN Base
- Makespan/pallet OUT (2) : from the arrival of the pallet to the last package dropped off on the OUT Base
- Computation time (3)

Validation:

- A **FlexSim model will be used by the jury** to check the results of the contestants (made public ~Nov. 2026)

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SUMMARY

INPUTS

Provided by the organisation

- Fixed configuration of the warehouse (Alleys, levels, bays, AGV fleets, geometry, IN/OUT Bases, etc.)
- Initial stock (SKU, redundancy, locations)
- Initial AGV positions
- Lists of IN/OUT missions for each instance

Management strategy using a combination of algorithms to achieve an optimal solution with the best possible performance metrics

OUTPUTS

Provided by the contestants

- A list of scheduled missions with their allocations for each instance (to be sent to FlexSim)
- A report explaining the implemented logic and justifying the choice of algorithms (respecting SOHOMA template)

FlexSim model
Provided by the organisation

KPIs

- Global makespan
- Makespan / pallet
- Computation time

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